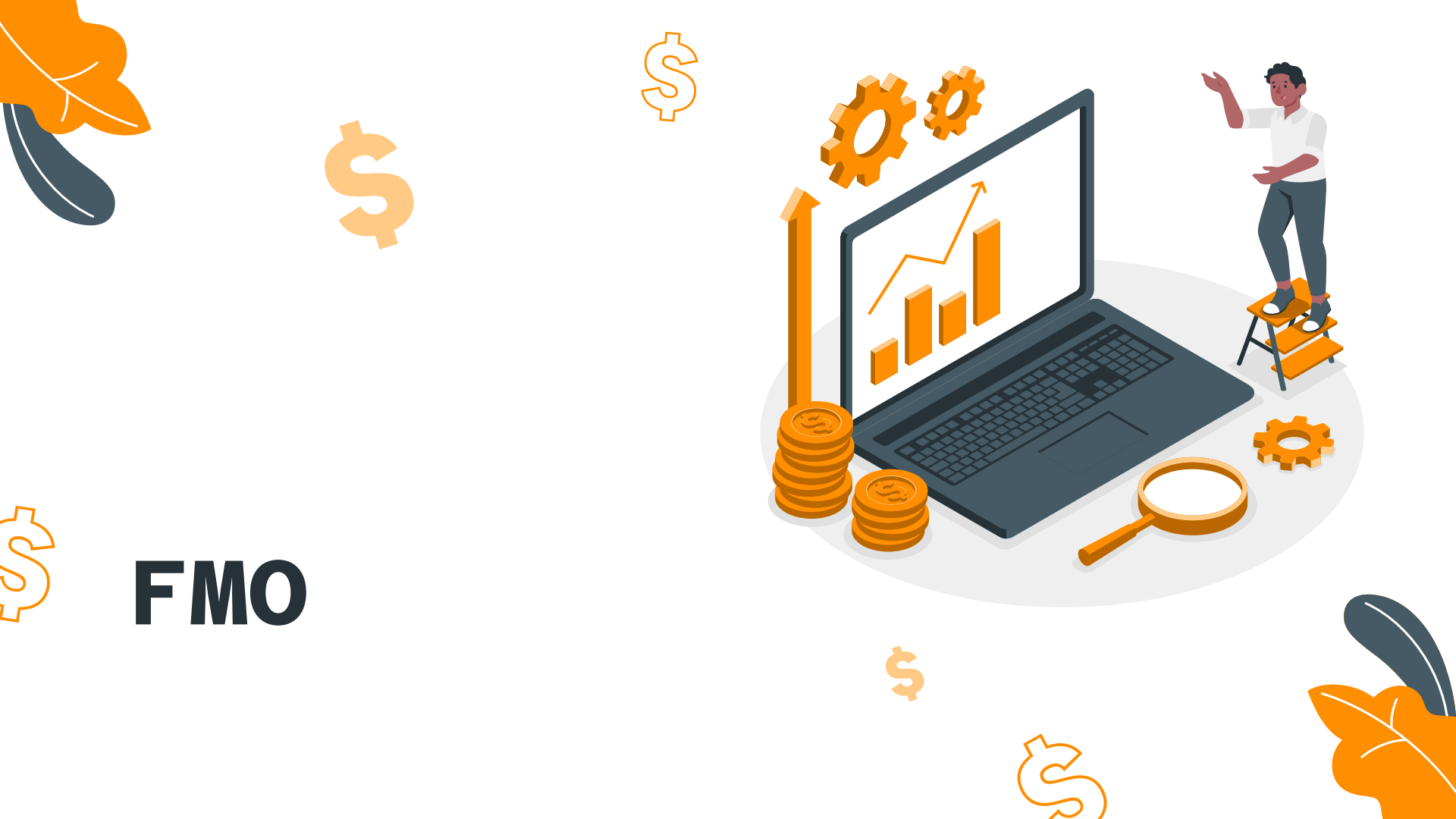




FMO



Project



Company Supreme Light is considering an investment in a new project that involves the development and launch of a new product line. The project requires an initial investment of \$500,000 in equipment and marketing expenses. The project is expected to generate cash flows over the next five years as follows:

- Year 1: \$100,000
 - Year 2: \$150,000
 - Year 3: \$200,000
 - Year 4: \$250,000
 - Year 5: \$300,000
- 

The company's cost of capital is 10%.

Should it invest in the new project based on the DCF analysis?





Project | \$

To determine whether the investment in the new project is feasible, we will perform a DCF analysis. The first step is to calculate the present value of each cash flow using the discount rate of 10%.





Project \$

Using the formula for present value:

$$PV = CF / (1+r)^n$$

Where:

- PV = Present Value
- CF = Cash Flow
- r = Discount Rate
- n = Number of Periods





Project



- Year 1: $PV = \$100,000 / (1 + 0.10)^1 = \$90,909.09$
 - Year 2: $PV = \$150,000 / (1 + 0.10)^2 = \$121,900.83$
 - Year 3: $PV = \$200,000 / (1 + 0.10)^3 = \$165,289.26$
 - Year 4: $PV = \$250,000 / (1 + 0.10)^4 = \$207,900.66$
 - Year 5: $PV = \$300,000 / (1 + 0.10)^5 = \$250,000.00$
- 
- 
- 



Project



Next, we sum up the present values of all cash flows:


$$\begin{aligned} NPV &= \sum PV - \text{Initial Investment} \\ NPV &= (\$90,909.09 + \$121,900.83 + \$165,289.26 + \$207,900.66 + \\ &\quad \$250,000.00) - \$500,000 \\ NPV &= \$835,999.84 - \$500,000 \\ NPV &= \$335,999.84 \end{aligned}$$




Risk and Return





Risk and Return

- Defining Risk and Return
- Using Probability Distributions to Measure Risk
- Attitudes Toward Risk
- Risk and Return in a Portfolio Context
- Diversification
- The Capital Asset Pricing Model (CAPM)
- Efficient Financial Markets





Risk and Return

Income received on an investment plus any **change in market price**, usually expressed as a percent of the **beginning market price** of the investment.


$$R = \frac{D_t + (P_t - P_{t-1})}{P_{t-1}}$$




Return –Case

The stock price for Stock A was \$10 per share 1 year ago. The stock is currently trading at \$9.50 per share and shareholders just received a \$1 dividend. What return was earned over the past year?





Risk and Return

$$R = \frac{\$1.00 + (\$9.50 - \$10.00)}{\$10.00} = 5\%$$





Risk and Return

$$R = \sum (R_i)(P_i)$$

R is the expected return for the asset,
 R_i is the return for the i^{th} possibility,
 P_i is the probability of that return occurring,
 n is the total number of possibilities.





Return



$$R = \sum_{i=1}^n (R_i)(P_i)$$

R is the expected return for the asset,
 R_i is the return for the i^{th} possibility,
 P_i is the probability of that return occurring,
 n is the total number of possibilities.





Return



Expected returns are based on the probabilities of possible outcomes

In this context, “expected” means average if the process is repeated many times

The “expected” return does not even have to be a possible return


$$E(R) = \sum_{i=1}^n p_i R_i$$




Return



Illustration 6: An investor expects future dividend from his investments in a company, and probability associated with the distribution of dividend at the expected level is given below. Compute expected return from such investments.

Possible return	40	30	35	52	45
Probability	0.60	0.55	0.40	0.70	0.76





Return



Solution:

Expected Returns

Possible Return (X_i)	Probability $p(X_i)$	Expected Return $X_i p(X_i)$
40	0.60	24
30	0.55	16.5
35	0.40	14
52	0.70	36.4
45	0.76	34.2
Sum of $X_i = 202$		$\Sigma X_i p(X_i) = 125.1$



Case: Expected Returns

- Suppose you have predicted the following returns for stocks Cefal and Toten in three possible states of nature. What are the expected returns?

○ State	Probability	Cefal	Toten
○ Boom	0.3	0.15	0.25
○ Normal	0.5	0.10	0.20
○ Recession	???	0.02	0.01

- $R_C = .3(.15) + .5(.10) + .2(.02) = .099 = 9.99\%$
- $R_T = .3(.25) + .5(.20) + .2(.01) = .177 = 17.7\%$